Lahore Urban Estates Analytics Warehouse

Business Questions

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The 14 analytical questions below were derived from the stakeholders' decisions, not from the source tables. Each one is traced to the fact table whose grain can answer it, the dimensions it slices by, its measures and the OLAP operation it needs. The brief requires at least 10.

1. The Questions

Collections (rent invoicing and payment)

Q1. What is the monthly collection rate by building?

Stakeholder: Finance head
Decision supported: Which buildings need collection follow-up this month
Answered by: fact_rent_invoice_line + fact_payment
Dimensions: dim_date (due), dim_unit
Measures: net_amount, payment_amount
OLAP operation: Drill-across on due date; ROLLUP Region → City → Building with GROUPING() labels

Q2. Which buildings have the highest arrears ratio?

Stakeholder: CEO, finance head
Decision supported: Where to escalate arrears recovery and review owner agreements
Answered by: fact_rent_invoice_line + fact_payment
Dimensions: dim_unit, dim_date (due)
Measures: net_amount, payment_amount
OLAP operation: Aggregate first, then Top-N (RANK)

Q3. How are overdue balances distributed across ageing buckets (0–30, 31–60, 61–90, 90+ days)?

Stakeholder: Finance head
Decision supported: Which balances to chase first and when to start legal notices
Answered by: fact_rent_invoice_line + fact_payment
Dimensions: dim_date (due), dim_charge_type
Measures: open balance
OLAP operation: Slice by bucket; drill bucket → building

Q4. Which tenants pay late repeatedly, and does late payment differ by employer sector at the time of payment?

Stakeholder: Finance head, building managers
Decision supported: Whether to ask for larger deposits or post-dated cheques from higher-risk segments
Answered by: fact_payment
Dimensions: dim_tenant (SCD2), dim_date (paid)
Measures: days_past_due (weighted), payment_amount
OLAP operation: Window functions (late months per tenant); slice by sector version

Q5. Which payment methods settle fastest after the due date?

Stakeholder: Finance head
Decision supported: Which payment channels to promote to tenants
Answered by: fact_payment
Dimensions: dim_payment_method
Measures: days_past_due, settlement_lag_days
OLAP operation: Roll-up Method → Channel group

Occupancy and leasing

Q6. How do occupancy and vacancy trend by city and unit type?

Stakeholder: Operations head
Decision supported: Where to focus leasing effort and adjust asking rents
Answered by: fact_unit_occupancy_daily
Dimensions: dim_date (snapshot), dim_unit
Measures: is_occupied
OLAP operation: CUBE City × Unit type; drill Year → Quarter → Month

Q7. Which unit types stay vacant longest?

Stakeholder: Leasing manager
Decision supported: Which unit types need price or furnishing changes
Answered by: fact_unit_occupancy_daily
Dimensions: dim_unit, dim_date
Measures: is_occupied (vacancy spells)
OLAP operation: Window functions (LAG) to find vacancy spells

Q8. What is the lease renewal rate, and which buildings lose tenants at renewal?

Stakeholder: Leasing manager
Decision supported: Where to run tenant retention before renewal dates
Answered by: fact_lease_event + dim_lease
Dimensions: dim_lease_event_type, dim_unit, dim_date (event)
Measures: event_count
OLAP operation: Ratio of counts by building; ROLLUP City → Building

Q13. Do vacancy incentives shorten vacancy, and which offer type works best?

Stakeholder: Leasing manager
Decision supported: Whether incentives are worth their cost, and which type to offer
Answered by: fact_vacancy_incentive_coverage + fact_unit_occupancy_daily
Dimensions: dim_incentive, dim_unit, dim_date
Measures: covered vs uncovered vacancy spell length
OLAP operation: Compare covered vs uncovered spells; slice by incentive type

Maintenance

Q9. What is the average time to resolve a request by category and vendor, and were low-rated vendors slower?

Stakeholder: Operations head
Decision supported: Which vendors to keep, retrain or replace
Answered by: fact_maintenance_request
Dimensions: dim_maintenance_category, dim_vendor (SCD2)
Measures: days_to_resolve
OLAP operation: Drill Category → Subcategory; slice by rating band in force

Q10. Which buildings breach the maintenance SLA most often?

Stakeholder: Operations head, building managers
Decision supported: Which buildings need more maintenance capacity or stricter vendor SLAs
Answered by: fact_maintenance_request
Dimensions: dim_unit, dim_maintenance_category
Measures: sla_breached_flag
OLAP operation: ROLLUP City → Building with GROUPING() labels

Q11. What is the maintenance cost per unit per month by building?

Stakeholder: Owner relations lead
Decision supported: How much maintenance reserve to budget per building
Answered by: fact_maintenance_request + dim_unit
Dimensions: dim_date (resolved), dim_unit
Measures: actual_cost
OLAP operation: Roll-up Month → Quarter → Fiscal year

Q14. Do leak and plumbing requests rise on heavy-rain days?

Stakeholder: Operations head
Decision supported: Whether to schedule pre-monsoon roof and drainage inspections
Answered by: fact_maintenance_request
Dimensions: dim_weather, dim_maintenance_category
Measures: request count
OLAP operation: Slice by rain band; drill Category → Subcategory

Owner reporting (cross-process)

Q12. What is the net operating income (rent collected − maintenance cost − LUE fee) per owner portfolio?

Stakeholder: CEO, owner relations lead
Decision supported: What to report to each owner and which portfolios are at risk of leaving LUE
Answered by: fact_payment + fact_maintenance_request
Dimensions: dim_owner, dim_date
Measures: payment_amount, actual_cost
OLAP operation: Drill-across on conformed Owner and Date

2. Grain Test: Why Only This Fact Can Answer It

Following Lab 05 Task 4.2, each group of questions was tested against every fact grain. The list records why the chosen fact is the only correct source.

- **Q1–Q3** — answered by Invoice line + payment facts. The occupancy snapshot holds only the current lease's arrears, so a departed tenant's debt would be missed.
- **Q4** — answered by fact_payment. Only the payment fact knows when money arrived; invoice lines record what was billed, not when it was paid.
- **Q6–Q7** — answered by fact_unit_occupancy_daily. Lease events record changes, not the state of every unit on every day, so vacancy days cannot be counted from them.
- **Q8** — answered by fact_lease_event. The occupancy snapshot shows a unit stayed occupied but cannot tell a renewal from a new tenant.
- **Q9–Q11** — answered by fact_maintenance_request. Only the accumulating snapshot holds every milestone date of a request on one row.
- **Q12** — answered by Two facts (drill-across). Rent and cost belong to different processes and meet only through conformed dim_owner and dim_date.
- **Q13** — answered by fact_vacancy_incentive_coverage. An offer that produced no lease leaves no row in any other fact; only the coverage fact records it.

3. Detail Versus Pre-Aggregation

The Gold marts store monthly summaries for Power BI, a MOLAP-style pre-aggregation (Lab 04 Section 9). Some questions need the detail that the marts discard:

- **“Which units were vacant during Eid week?”** can be answered from fact_unit_occupancy_daily but not from mart_occupancy_monthly, because the day was aggregated away.
- **Q7 vacancy spells** need consecutive days per unit, which only the daily snapshot keeps.
- **Q1** will be answered both ways in Phase 3 to compare joins, query-time aggregation and the detail available (ROLAP vs MOLAP vs HOLAP).

4. Example SQL

These sketches show how five questions run on the physical model. Each query was executed against the PostgreSQL 16 schema in sql/physical_model.sql. Subtotal rows are always labelled with GROUPING(), so a manager never has to interpret a bare NULL (Lab 04 Tasks 6.2 and 7.4).

Q1: Collection rate by Region → City → Building with labelled subtotals (ROLLUP, drill-across)

Invoice lines and payments are combined at the same conformed grain (unit, due date) before one ROLLUP, so billed and collected amounts are never double-counted.

```
WITH lines AS (                       -- drill-across: two facts, one conformed grain
    SELECT i.unit_key, i.due_date_key, i.net_amount AS billed, 0::numeric AS collected
    FROM lue_dw.fact_rent_invoice_line i
    UNION ALL
    SELECT p.unit_key, p.due_date_key, 0, p.payment_amount
    FROM lue_dw.fact_payment p
)
SELECT CASE WHEN GROUPING(u.region) = 1        THEN 'Grand total'
            WHEN GROUPING(u.city) = 1          THEN 'Region subtotal'
            WHEN GROUPING(u.building_name) = 1 THEN 'City subtotal'
            ELSE 'Building' END                  AS row_type,
       u.region, u.city, u.building_name,
       SUM(l.billed)                             AS billed_pkr,
       SUM(l.collected)                          AS collected_pkr,
       ROUND(100 * SUM(l.collected) / NULLIF(SUM(l.billed), 0), 1) AS collection_rate_pct
FROM lines l
JOIN lue_dw.dim_unit u ON u.unit_key = l.unit_key
JOIN lue_dw.dim_date d ON d.date_key = l.due_date_key
WHERE d.year_no = 2026 AND d.month_no = 3
GROUP BY ROLLUP (u.region, u.city, u.building_name)
ORDER BY u.region NULLS LAST, u.city NULLS LAST, u.building_name NULLS LAST;
```

Q4: Late payment by employer sector at the time of payment (SCD Type 2 point-in-time)

fact_payment.tenant_key already points at the tenant version valid on the paid date, so a tenant who changed sector is counted under the sector they had when they paid. days_past_due is non-additive, so it is amount-weighted.

```
SELECT t.employer_sector,
       COUNT(DISTINCT p.payment_reference)  AS payments,
       ROUND(SUM(p.days_past_due * p.payment_amount)
             / NULLIF(SUM(p.payment_amount), 0), 1) AS weighted_days_past_due,
       ROUND(100 * SUM(CASE WHEN p.days_past_due > 0 THEN p.payment_amount ELSE 0 END)
             / NULLIF(SUM(p.payment_amount), 0), 1) AS pct_amount_paid_late
FROM lue_dw.fact_payment p
JOIN lue_dw.dim_tenant t ON t.tenant_key = p.tenant_key   -- version valid on the paid date
JOIN lue_dw.dim_date   d ON d.date_key   = p.paid_date_key
WHERE d.fiscal_year_label = 'FY2025-26'
  AND NOT p.is_reversal
GROUP BY t.employer_sector
ORDER BY weighted_days_past_due DESC;
```

Q6: Occupancy rate City × Unit type with every subtotal labelled (CUBE)

is_occupied is semi-additive: summed across units and divided by unit-days, never summed across days on its own.

```
SELECT CASE WHEN GROUPING(u.city) = 1 AND GROUPING(u.unit_type) = 1 THEN 'Grand total'
            WHEN GROUPING(u.city) = 1      THEN 'Unit type total (all cities)'
            WHEN GROUPING(u.unit_type) = 1 THEN 'City total (all unit types)'
            ELSE 'City x unit type' END     AS row_type,
       u.city, u.unit_type,
       ROUND(100.0 * SUM(o.is_occupied) / COUNT(*), 1) AS occupancy_rate_pct
FROM lue_dw.fact_unit_occupancy_daily o
JOIN lue_dw.dim_unit u ON u.unit_key = o.unit_key
JOIN lue_dw.dim_date d ON d.date_key = o.snapshot_date_key
WHERE d.fiscal_year_label = 'FY2025-26'
GROUP BY CUBE (u.city, u.unit_type)
ORDER BY GROUPING(u.city), GROUPING(u.unit_type), u.city, u.unit_type;
```

Q7: Average vacancy spell by unit type (window functions on the daily snapshot)

Consecutive vacant days are grouped into spells with the gaps-and-islands technique; this question cannot be answered from a monthly aggregate because the day-level sequence is lost.

```
WITH days AS (
    SELECT u.unit_nk, u.unit_type, d.full_date, o.is_occupied,
           d.full_date - ROW_NUMBER() OVER (PARTITION BY u.unit_nk, o.is_occupied
                                            ORDER BY d.full_date)::int AS spell_id
    FROM lue_dw.fact_unit_occupancy_daily o
    JOIN lue_dw.dim_unit u ON u.unit_key = o.unit_key
    JOIN lue_dw.dim_date d ON d.date_key = o.snapshot_date_key
), spells AS (
    SELECT unit_nk, unit_type, spell_id, COUNT(*) AS vacant_days
    FROM days
    WHERE is_occupied = 0
    GROUP BY unit_nk, unit_type, spell_id
)
SELECT unit_type,
       COUNT(*)                                    AS vacancy_spells,
       ROUND(AVG(vacant_days), 1)                  AS avg_vacant_days,
       RANK() OVER (ORDER BY AVG(vacant_days) DESC) AS rank_longest
FROM spells
GROUP BY unit_type;
```

Q12: Net operating income per owner (drill-across on conformed Owner and Date)

Rent and maintenance cost come from different processes; each is aggregated to the owner first and only then combined. The management fee rate is a stated assumption (8% of rent collected).

```
WITH rent AS (
    SELECT p.owner_key, SUM(p.payment_amount) AS rent_collected
    FROM lue_dw.fact_payment p
    JOIN lue_dw.dim_date d        ON d.date_key        = p.paid_date_key
    JOIN lue_dw.dim_charge_type c ON c.charge_type_key = p.charge_type_key
    WHERE d.fiscal_year_label = 'FY2025-26' AND c.charge_category = 'Rent'
    GROUP BY p.owner_key
), cost AS (
    SELECT m.owner_key, SUM(m.actual_cost) AS maintenance_cost
    FROM lue_dw.fact_maintenance_request m
    JOIN lue_dw.dim_date d ON d.date_key = m.resolved_date_key
    WHERE d.fiscal_year_label = 'FY2025-26'
    GROUP BY m.owner_key
)
SELECT o.owner_name,
       COALESCE(r.rent_collected, 0)                   AS rent_collected,
       COALESCE(c.maintenance_cost, 0)                 AS maintenance_cost,
       ROUND(0.08 * COALESCE(r.rent_collected, 0), 2)  AS lue_fee,
       ROUND(0.92 * COALESCE(r.rent_collected, 0) - COALESCE(c.maintenance_cost, 0), 2)
                                                       AS net_operating_income
FROM lue_dw.dim_owner o
LEFT JOIN rent r ON r.owner_key = o.owner_key
LEFT JOIN cost c ON c.owner_key = o.owner_key
WHERE o.owner_key > 0
ORDER BY net_operating_income DESC;
```